Reliability Assessment of hBN-Encapsulated TMD Devices from FETs to Inverters

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***Abstract*—This document is a model and instructions for LaTeX. This and the IEEEtran.cls file define the components of your paper [title, text, heads, etc.]. *CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract.**

***Index Terms*—component, formatting, style, styling, insert**

I. Introduction

As device miniaturization continues, the performance of traditional logic devices based on bulk semiconductors becomes increasingly constrained, either by the exacerbation of short-channel effects in field-effect transistors (FETs) when the channel thickness is not scaled accordingly, or by the associated mobility reduction when it is [1]. To overcome this inherent trade-off of bulk channels, two-dimensional (2D) materials, whose carrier mobility exhibits weak thickness dependence, have attracted significant attention. Among them, transition-metal dichalcogenides (TMDs) offer suitable bandgaps and currently represent the backbone of active logic devices based on 2D materials, with MoS_2 being the most mature option for realizing n-channel FETs [2].

However, important limitations arise with the use of 2D materials. While their dangling-bond-free interfaces mitigate the mobility drop for scaled devices, the absence of covalent bonding at the oxide–channel interface leads to increased instability compared to the well-established SiO_2/Si interface [3]. This instability typically originates from charge transfer and transport processes through the gate stack, manifests during device operation, and entails shifts in key device parameters (most notably the threshold voltage), thereby compromising the functionality and performance of the resulting logic circuits [4].

To mitigate the aforementioned instabilities, different gate stack concepts and optimizations have been proposed to reduce charge transfer without compromising device performance. A promising approach involves the use of 2D insulators to achieve well-passivated interfaces with the channel, thereby enhancing device stability, while enabling the construction of van der Waals heterostructures to tune device properties [5]. Among the candidate materials, hexagonal boron nitride (hBN) is the most ubiquitous; however, its low dielectric constant limits its use as single insulating material in deeply scaled devices due to the expected high gate leakage [6]. However, recent studies propose the use of hBN layers to encapsulate a bulk gate oxide, thereby improving device stability at the expense of increased fabrication complexity [7].

To benchmark this and other proposed approaches, it is essential to quantify the detrimental instability effects, which can be observed either under varying device conditions, as in classical hysteresis and bias temperature instability (BTI) measurements where the gate voltage is altered, or under constant conditions, as in low-frequency noise measurements. In particular, the former have proven to be straightforward and highly effective techniques for assessing device reliability [8]. However, inaccurate conclusions may be drawn if key experimental parameters and essential characteristics of the devices under test are overlooked. For instance, while the time required to perform the up- and down-sweeps in hysteresis measurements is rarely reported, it has been shown to have a significant impact on the underlying physical mechanisms of hysteresis and the resulting measurements [9]. Similarly, as the impact of most degradation mechanisms depends on insulator thickness, it becomes essential to normalize both hysteresis and BTI shifts with respect to thickness to enable accurate comparisons between samples.

In this work, a comprehensive reliability assessment of hBN-encapsulated MoS_2 FETs is presented, with hysteresis and BTI measurements constituting the core of the analysis and providing complementary insights. Special emphasis is placed on the number of analyzed devices to capture variability in instability behavior, as well as on key measurement parameters, such as the sweep time and the drain voltage in hysteresis measurements. Additionally, this analysis is extended to characterize depletion-load nMOS inverters realized with the previous FETs. The analysis is further extended to characterize depletion-load nMOS inverters based on these hBN-encapsulated FETs, enabling a straightforward evaluation of the reliability of this platform for deeply scaled logic applications. The rest of the manuscript is organized as follows...

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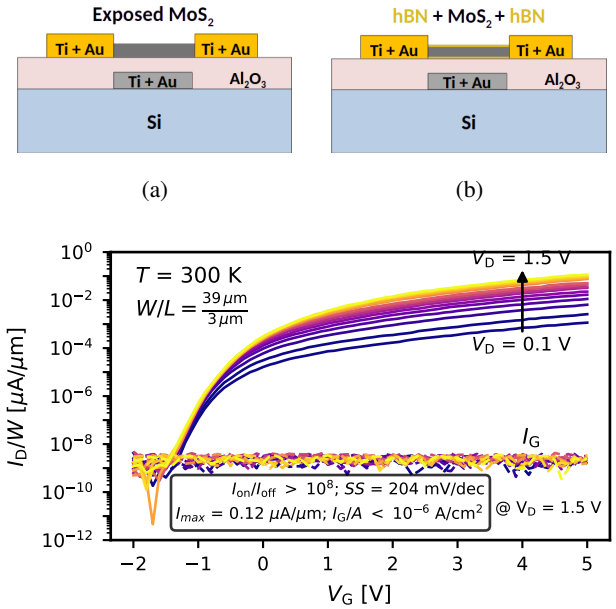


(a) (b)

(c)

Fig. 1. **Cross-section schematics**: (a) exposed and (b) hBN-encapsulated MoS_2 FETs. (c) **Performance of the fabricated devices**: $I_{D(G)}$-V_G curves for different drain voltages and the main device metrics.

II. Device Fabrication and Performance

III. Reliability of FETs

A. Hysteresis Characterization

B. Bias Temperature Instability

IV. Reliability of Inverters

V. Conclusions

Acknowledgment

The preferred spelling of the word "acknowledgment" in America is without an "e" after the "g". Avoid the stilted expression "one of us (R. B. G.) thanks ...". Instead, try "R. B. G. thanks...". Put sponsor acknowledgments in the unnumbered footnote on the first page.

References

[1] Y. Liu, X. Duan, H.-J. Shin, S. Park, Y. Huang, and X. Duan, "Promises and prospects of two-dimensional transistors," en, *Nature*, vol. 591, no. 7848, pp. 43–53, Mar. 2021.

[2] S. Das *et al.*, "Transistors based on two-dimensional materials for future integrated circuits," en, *Nature Electronics*, vol. 4, no. 11, pp. 786–799, Nov. 2021.

[3] H.-Y. Lan *et al.*, "Reliability of high-performance monolayer MoS2 transistors on scaled high-k HfO2," en, *npj 2D Materials and Applications*, vol. 9, no. 1, p. 5, Jan. 2025.

[4] Mulong Luo, Runsheng Wang, Shaofeng Guo, Jing Wang, Jibin Zou, and Ru Huang, "Impacts of Random Telegraph Noise (RTN) on Digital Circuits," en, *IEEE Transactions on Electron Devices*, vol. 62, no. 6, pp. 1725–1732, Jun. 2015.

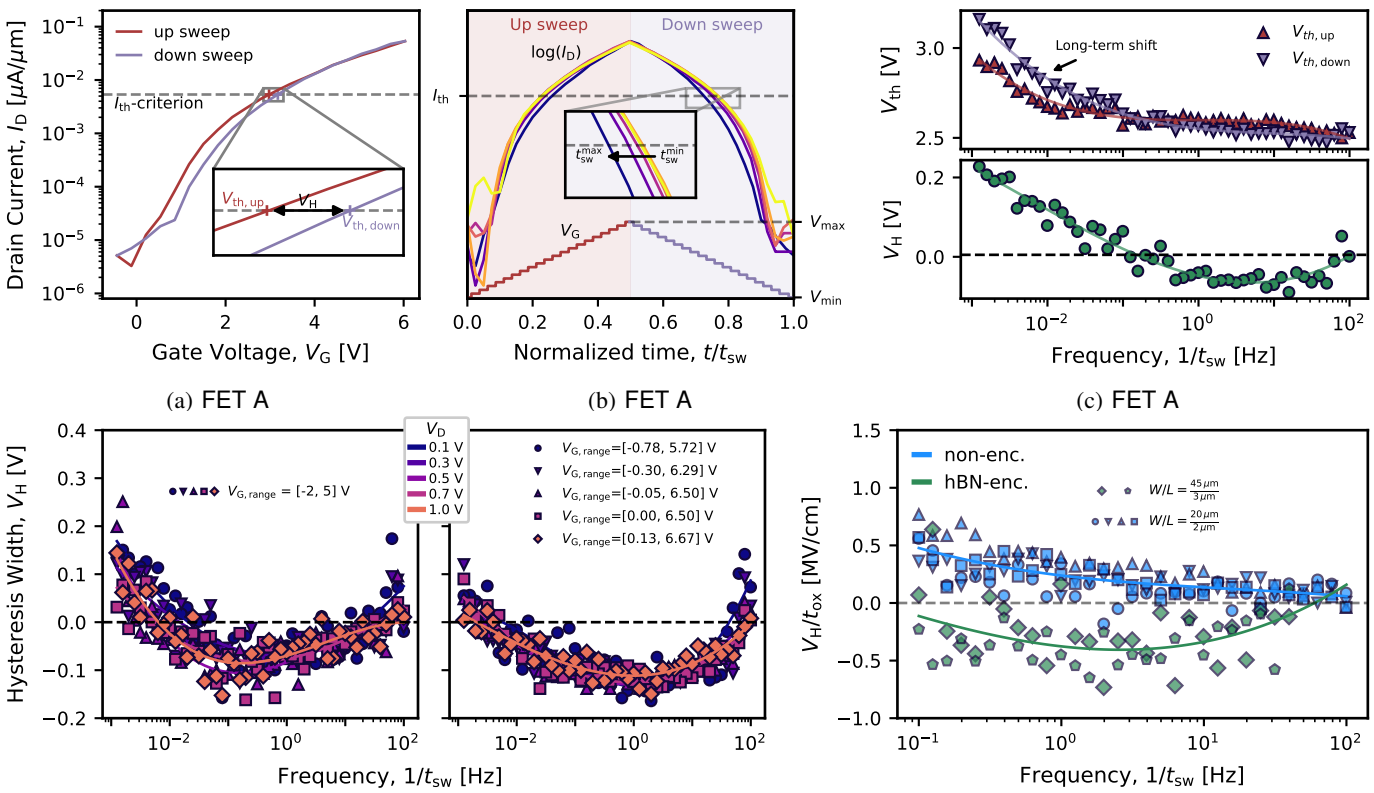
[5] A. K. Geim and I. V. Grigorieva, "Van der Waals heterostructures," en, *Nature*, vol. 499, no. 7459, pp. 419–425, Jul. 2013.

[6] T. Knobloch *et al.*, "The performance limits of hexagonal boron nitride as an insulator for scaled CMOS devices based on two-dimensional materials," en, *Nature Electronics*, vol. 4, no. 2, pp. 98–108, Feb. 2021.

[7] A. Piacentini *et al.*, "Stable Al2O3 Encapsulation of MoS2-FETs Enabled by CVD Grown h-BN," en, *Advanced Electronic Materials*, vol. 8, no. 9, p. 2 200 123, 2022.

[8] A. Provias *et al.*, "Reliability Assessment of Double-Gated Wafer-Scale MoS_2 Field Effect Transistors through Hysteresis and Bias Temperature Instability Analyses," en, in *2023 International Electron Devices Meeting (IEDM)*, San Francisco, CA, USA: IEEE, Dec. 2023, pp. 1–4.

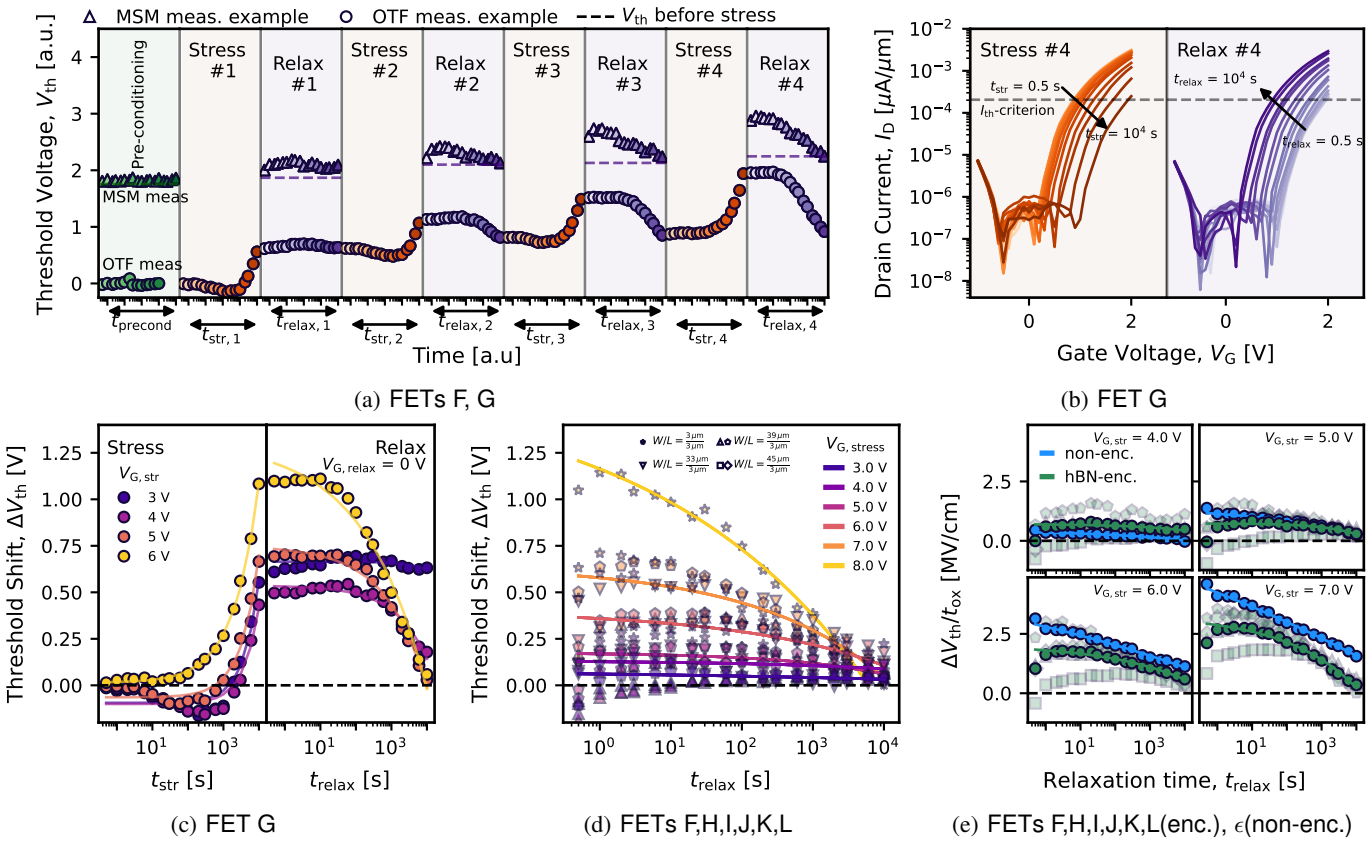
[9] A. Karl *et al.*, "A standardized approach to characterize hysteresis in 2D-materials-based transistors for stability benchmarking and performance projection," en, *Nature Communications*, vol. 17, no. 1, p. 171, Nov. 2025.



(a) FET A (b) FET A (c) FET A

(d) FETs B, C (e) FETs D, E(enc.), $\alpha, \beta, \gamma, \delta$(non-enc.)

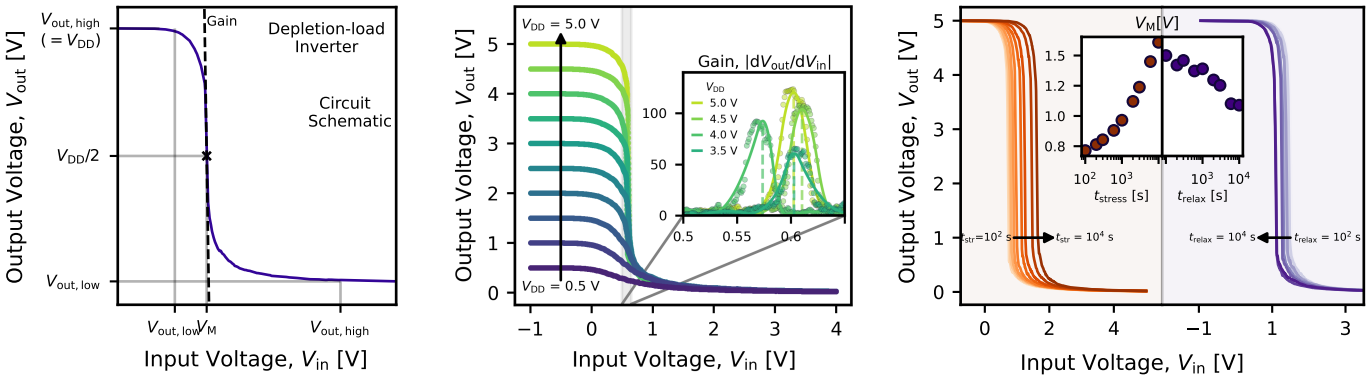
Fig. 2. (a) **Typical hysteresis curve**: the hysteresis width, V_H, is defined as the difference between the threshold voltages extracted during the up and down sweeps, i.e., $V_H = V_{th,up} - V_{th,down}$ evaluated at a constant current criterion I_{th}. (b) **Hysteresis dependence on the gate voltage sweep rate**: $\log(I_D)$ and V_G are shown as a function of the normalized sweep time, t/t_{sw}, for different sweep times, t_{sw}. (c) **Hysteresis-$1/t_{sw}$ curves**: Threshold voltages, $V_{th,up}$ and $V_{th,down}$, and hysteresis width, V_H, are shown as a function of the sweep frequency, $1/t_{sw}$. (d) **Hysteresis for varying V_D**: The hysteresis is independent from the applied drain bias, as evaluated on two different devices. More consistent hysteresis results are obtained when using an adaptive scheme for the gate voltage range targeting the same overdrive and the same sweep range in the subthreshold (right), as compared to a fixed gate voltage range (left). (e) **Hysteresis on encapsulated vs. non-encapsulated devices**: The hysteresis width on encapsulated devices is negative (counter-clockwise) and on non-encapsulated devices it was of a similar magnitude but positive (clockwise) evaluated at $V_D = 0.1V$.



(a) FETs F, G (b) FET G

(c) FET G (d) FETs F,H,I,J,K,L (e) FETs F,H,I,J,K,L(enc.), ϵ(non-enc.)

Fig. 3. (a) **Typical BTI experiments**: The device under test undergoes alternating periods of stress and recovery bias conditions.



(a) Inverter A (b) Inverter A (c) Inverter B

Fig. 4. (a) Example of a TDDB curve and the corresponding time evolution of the gate voltage. (b) Comparison of the $I_{ds} - V_{gs}$ curves for the RO stage and the asymmetric stage.